What is Autopilot

Windows Autopilot helps us set up and prepare new Windows devices for our organization without reimaging them. Imagine we buy a batch of new Windows PCs. Traditionally, IT would wipe each one and install either a fresh copy of Windows or our own custom image, which takes a lot of time and effort.

With Autopilot, we use the Windows that's already installed by the manufacturer. We register the devices' hardware hashes with Microsoft, and when a user turns on a device and goes through Windows setup (OOBE), Autopilot recognizes it as ours. It customizes the setup screens, joins the device to Microsoft Entra ID, and enrolls it in Intune. Intune then applies our security settings, configurations, and apps, and can remove software we don't need.

This means devices can be shipped directly from the vendor to the user, and IT doesn't have to touch each machine. Autopilot can also be used to reset and prepare existing devices again.

Autopilot also lets us control the OOBE process. We can hide screens the user doesn't need, such as the license terms (EULA), privacy settings, and the options to change account, so the user must sign in with their work account. We can also choose whether the user's Entra account becomes a standard user or a local administrator on the device.

We can also name devices automatically using a naming template (up to 15 characters), for example `NYC-%RAND:5%`. When the user signs in during OOBE, the device is associated with that user and they become its primary user in Intune.

How it works:

1. **The device turns on and goes to OOBE.** The user picks a region and keyboard and connects to the internet.
2. **The device asks Microsoft who it belongs to.** It sends its hardware hash to the Autopilot service. If the hash is registered, Microsoft tells it which tenant it belongs to.
3. **The device downloads the Autopilot deployment profile.** The profile sets how OOBE should look and behave: join type (Entra join or hybrid), pages to skip, device name template, and standard or admin user.
4. **The user authenticates.** They sign in with their work credentials on your company branding screen.
5. **The device joins Entra ID and then auto-enrolls in Intune.** This needs the MDM user scope turned on and an Intune license for the user.
6. **The Enrollment Status Page (ESP) runs.** Intune applies the policies and required apps targeted to the device or user (through their groups), and the user reaches the desktop when they finish.

Capture the hardware hash

Registration comes first, because when a device starts up and asks Microsoft who it belongs to, Microsoft looks for its **hardware hash** to find the right tenant. Devices from an OEM or reseller can be registered for you at purchase, but for a lab or an existing device, we capture the hash ourselves.

To capture it, we use a PowerShell script called **Get-WindowsAutopilotInfo**, which we run **on the device itself**.

1. Open PowerShell as administrator. (In OOBE, press **Shift+F10** and type `powershell`.)
2. Set the execution policy for this session to Bypass or Unrestricted:

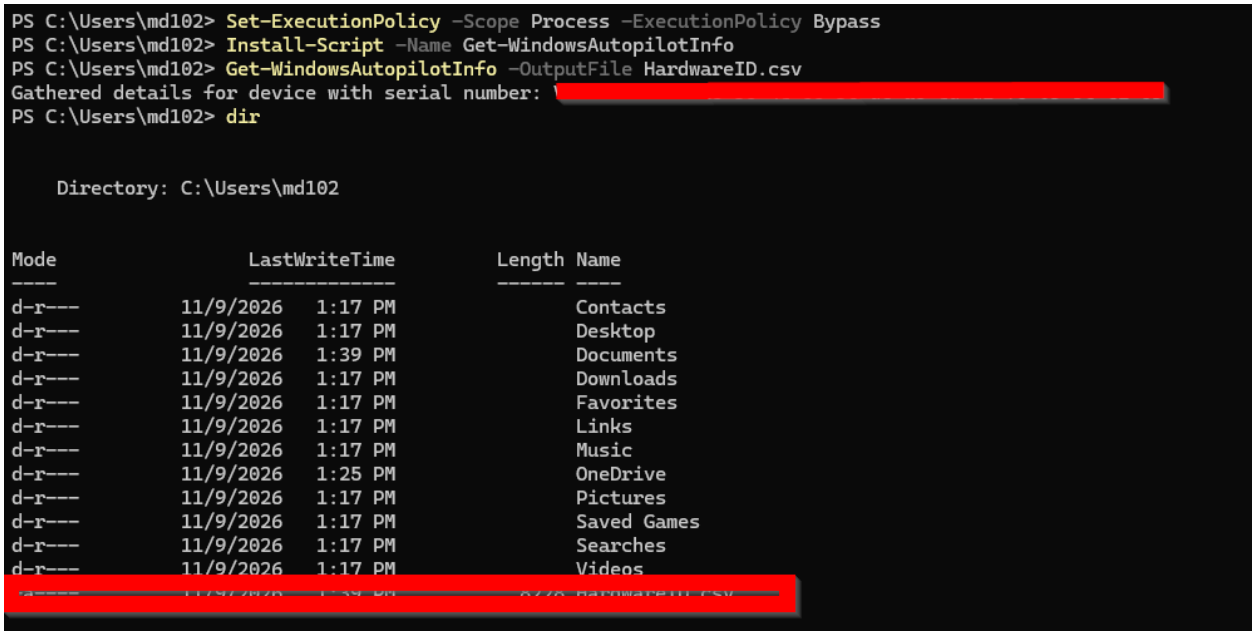
```
Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass
```

3. Install the script:

```
Install-Script -Name Get-WindowsAutopilotInfo
```

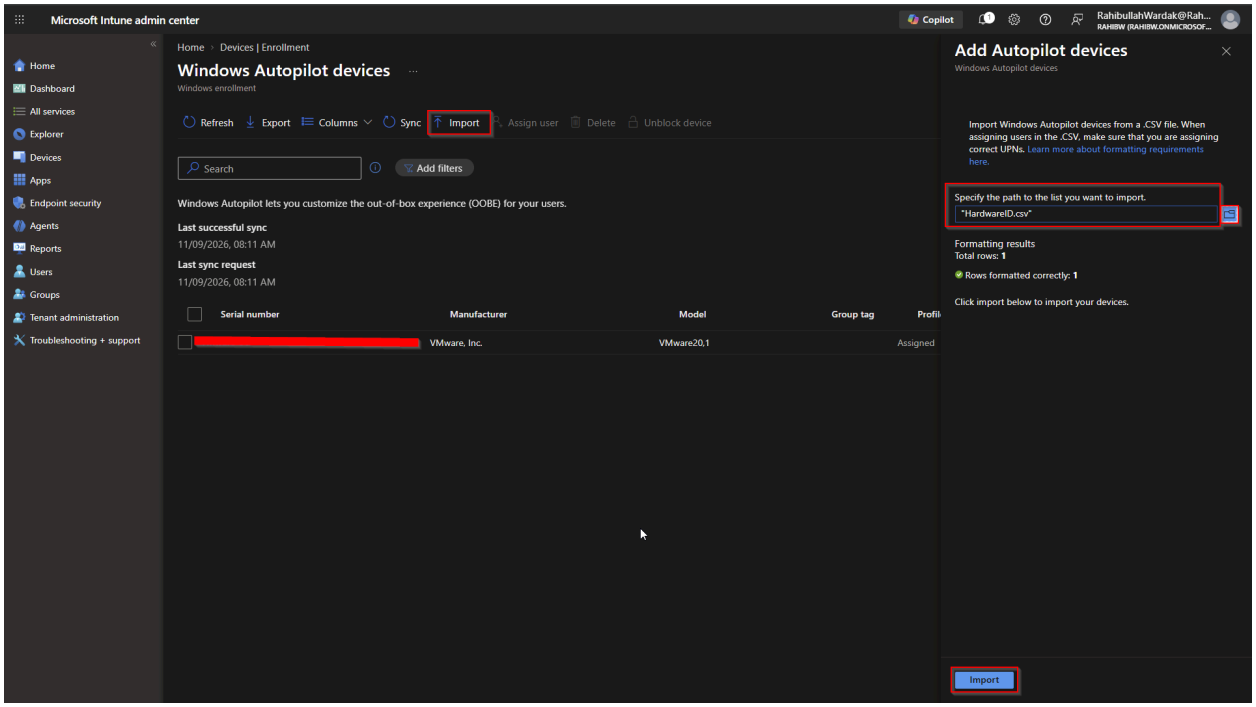
4. Export the hash to a CSV:

```
Get-WindowsAutopilotInfo -OutputFile HardwareID.csv
```



Registering the hash in your tenant tells Autopilot that this device belongs to your organization.

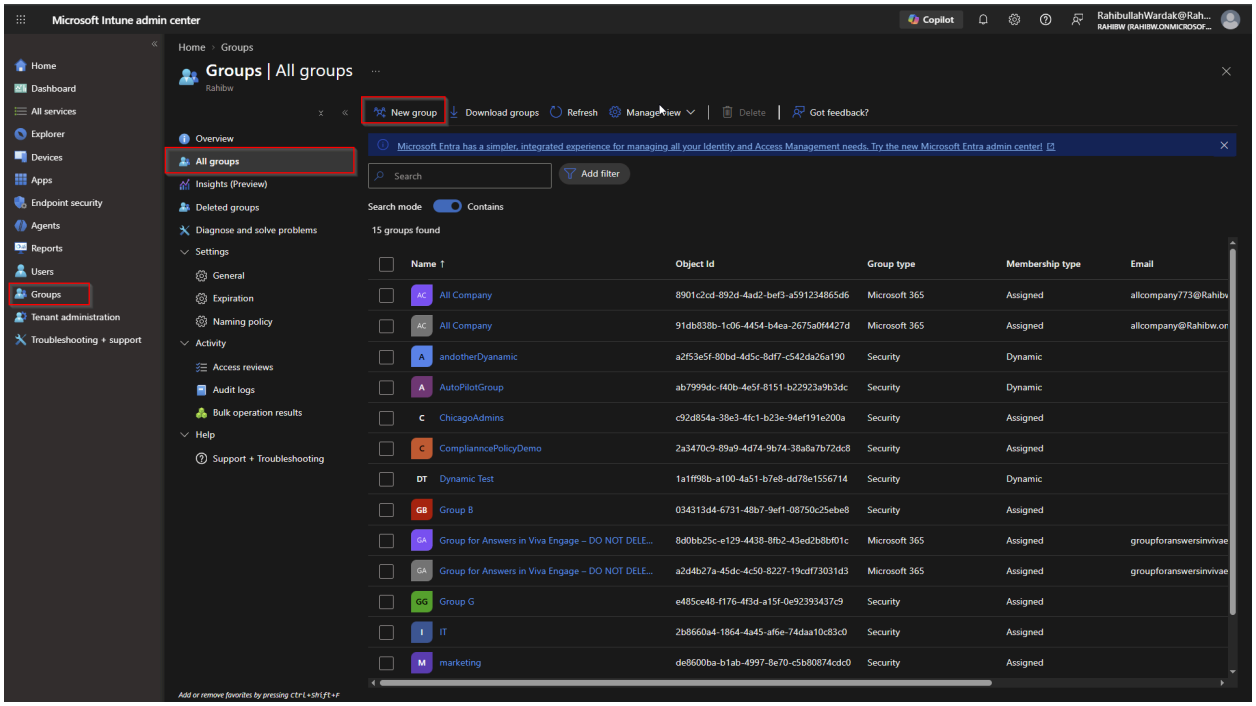
1. Sign in to the **Microsoft Intune admin center**.
2. Go to **Devices → Enrollment → Windows Autopilot → Devices**.
3. Select **Import**.
4. Choose the `HardwareID.csv` file you exported.
5. Select **Import** and wait for it to finish. This can take several minutes.
6. Select **Sync** and refresh the list until the device appears.



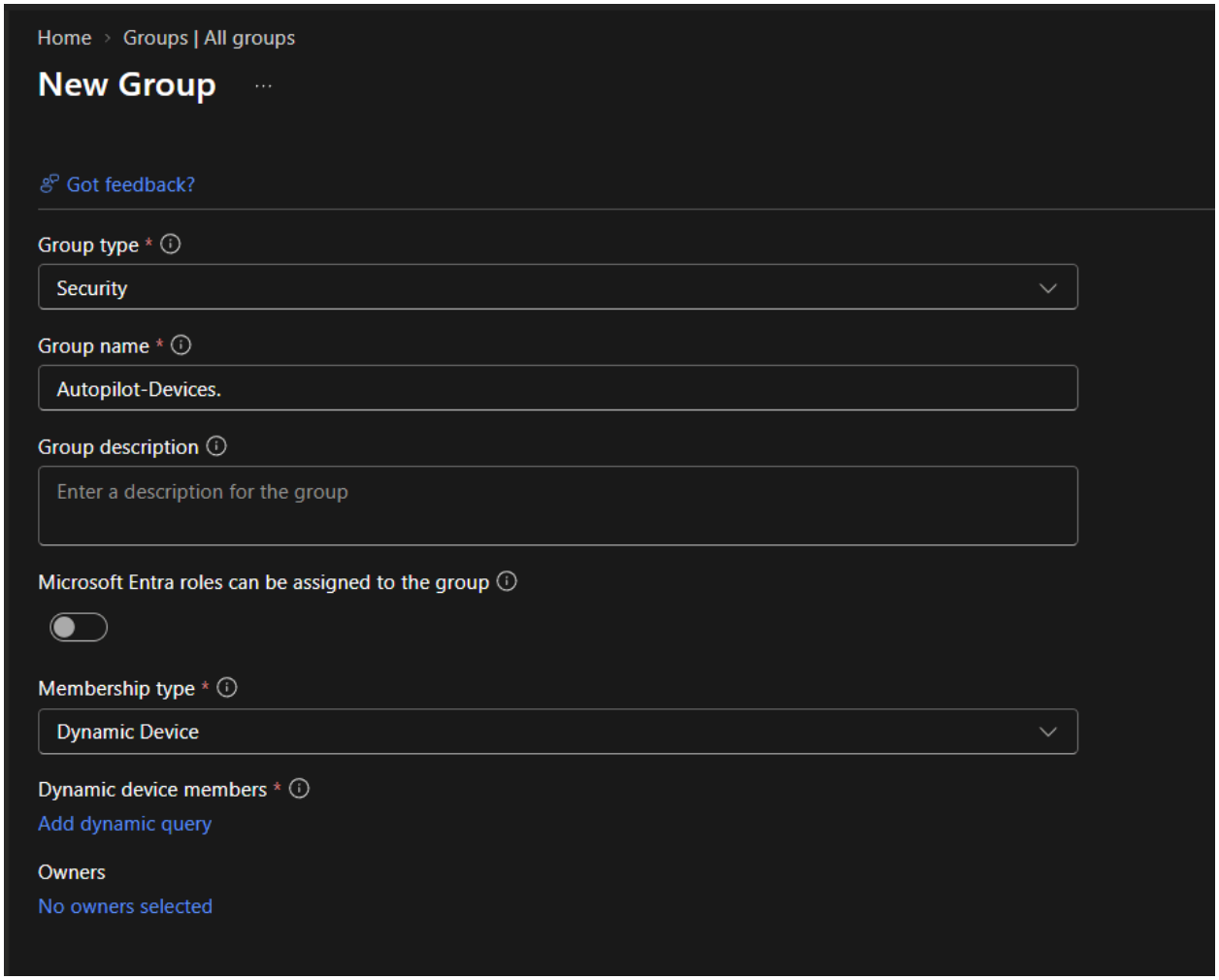
Create a dynamic group for Autopilot devices

A dynamic group automatically adds devices that match a rule, so each registered device doesn't have to be added by hand.

1. In the **Microsoft Entra admin center/Intune** , go to **Groups → All groups → New group**.



2. Set **Group type** to **Security**.
3. Enter a group name, for example `Autopilot-Devices`.
4. Set **Membership type** to **Dynamic Device**.



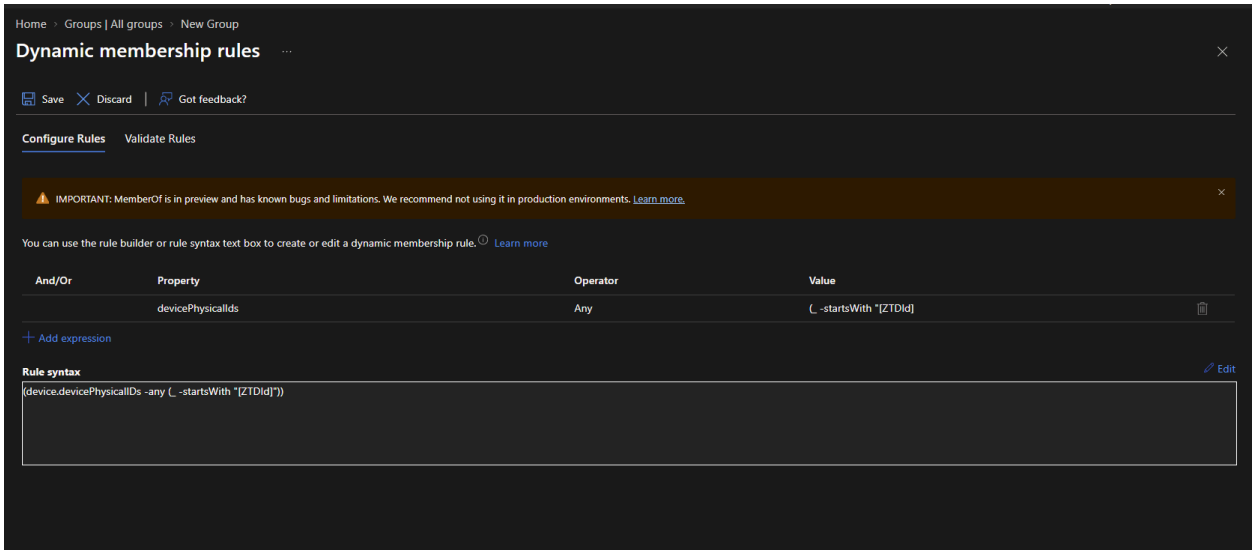
5. Select **Add dynamic query**, switch to **Rule syntax**, and enter the rule below.

To add every device registered in Autopilot:

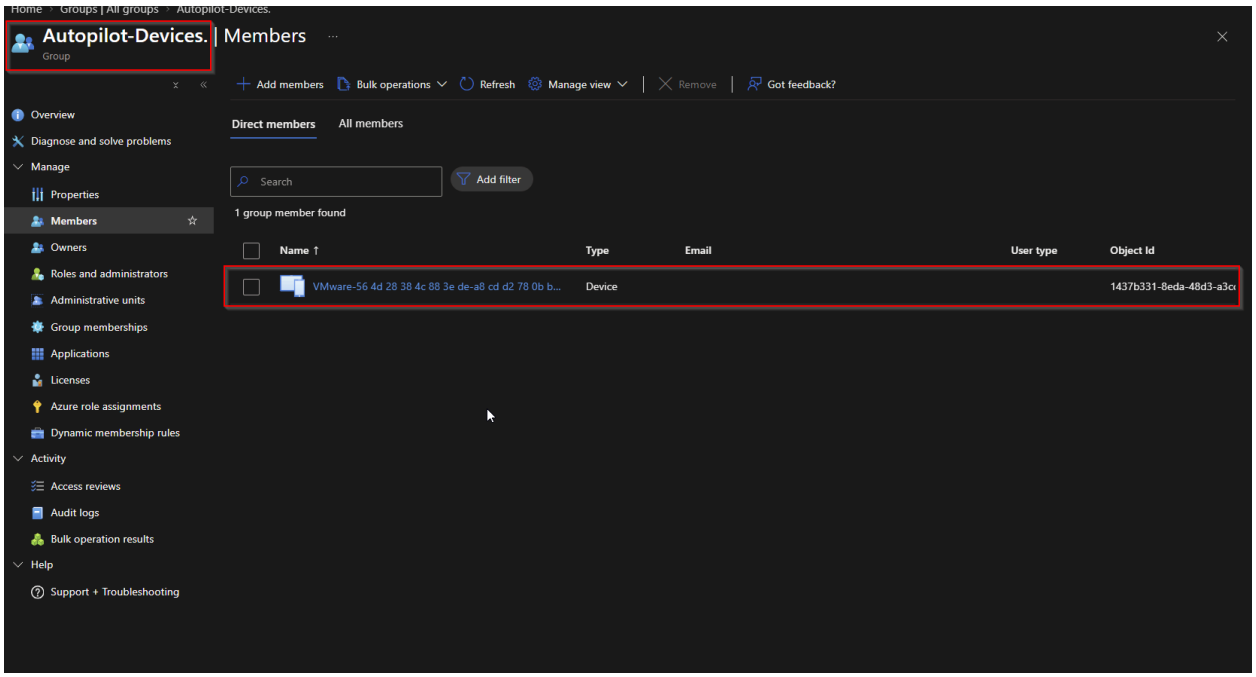
```
(device.devicePhysicalIDs -any (_ -startsWith "[ZTDId]"))
```

To add only devices with a specific Group Tag (replace `LabVM` with your tag):

```
(device.devicePhysicalIds -any (_ -eq "[OrderID]:LabVM"))
```



6. Select **Save**, then **Create**.
7. Open the group and check **Members**. The device can take several minutes to appear.



Note: Dynamic groups require Microsoft Entra ID P1.

Create an Autopilot deployment profile and assign it to the dynamic group

An Autopilot deployment profile defines how the out-of-box experience (OOBE) behaves on the devices it's assigned to. It controls settings such as the join type, whether the user can change account options, whether the user is a local administrator, and whether the privacy settings and license terms (EULA) are shown.

Autopilot deployment profile settings explained:

Deployment mode: User Driven: The user signs in during OOBE with their work account, and the device is set up for that user. The alternative, self-deploying, needs no user sign-in and is used for kiosks and shared devices.

Join to Microsoft Entra ID as: Microsoft Entra joined: Sets how the device joins your organization. Entra joined makes it a cloud-only device. Hybrid joined also joins it to on-premises Active Directory and needs extra setup (Entra Connect and a domain controller connection).

Microsoft Software License Terms: Hide : Skips the Windows license agreement (EULA) page and accepts it on the user's behalf. This removes a step and keeps users from declining it.

Privacy settings: Hide : Skips the privacy settings page (location, diagnostics, and so on), so the user doesn't have to choose. The organization sets these through Intune policy instead.

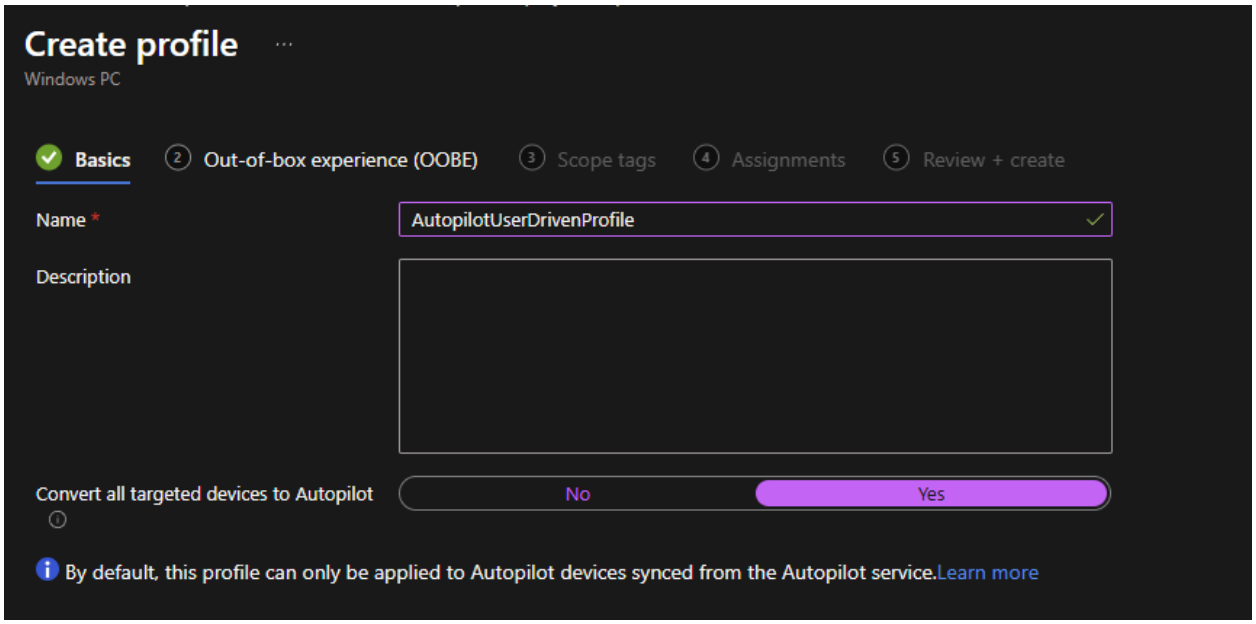
Hide change account options: Hide :Hides the option to use a different account on the sign-in and error screens. It keeps the user on the company sign-in instead of a personal Microsoft account.

User account type: Standard :The user who signs in gets a standard account instead of local administrator rights. Standard is the safer choice because users can't install unapproved software or change system settings. *Administrator* gives them full local rights.

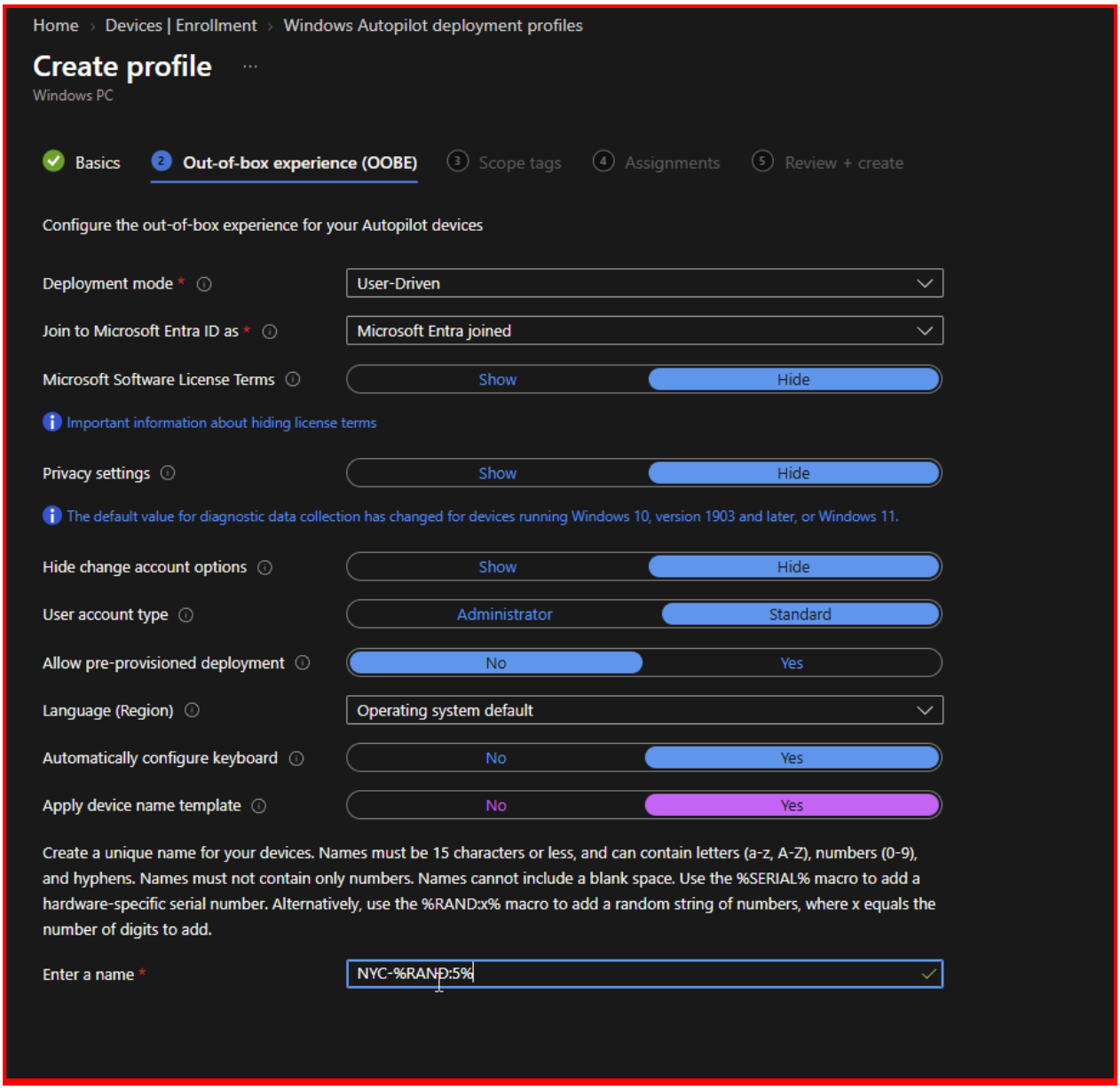
Apply device name template : Names devices automatically using a pattern, for example `NYC-%RAND:5%`. It saves manual renaming and keeps names consistent. Names are limited to 15 characters.

Allow pre-provisioned deployment: When this is set to **Yes**, IT (or a vendor) can set up the device before it reaches the user. The technician starts Autopilot, and the device installs its apps, policies, and settings, which is the device-based portion of setup. The device is then handed to the user, who only has to sign in to finish. This makes the user's first-time setup much faster, because most of the work is already done.

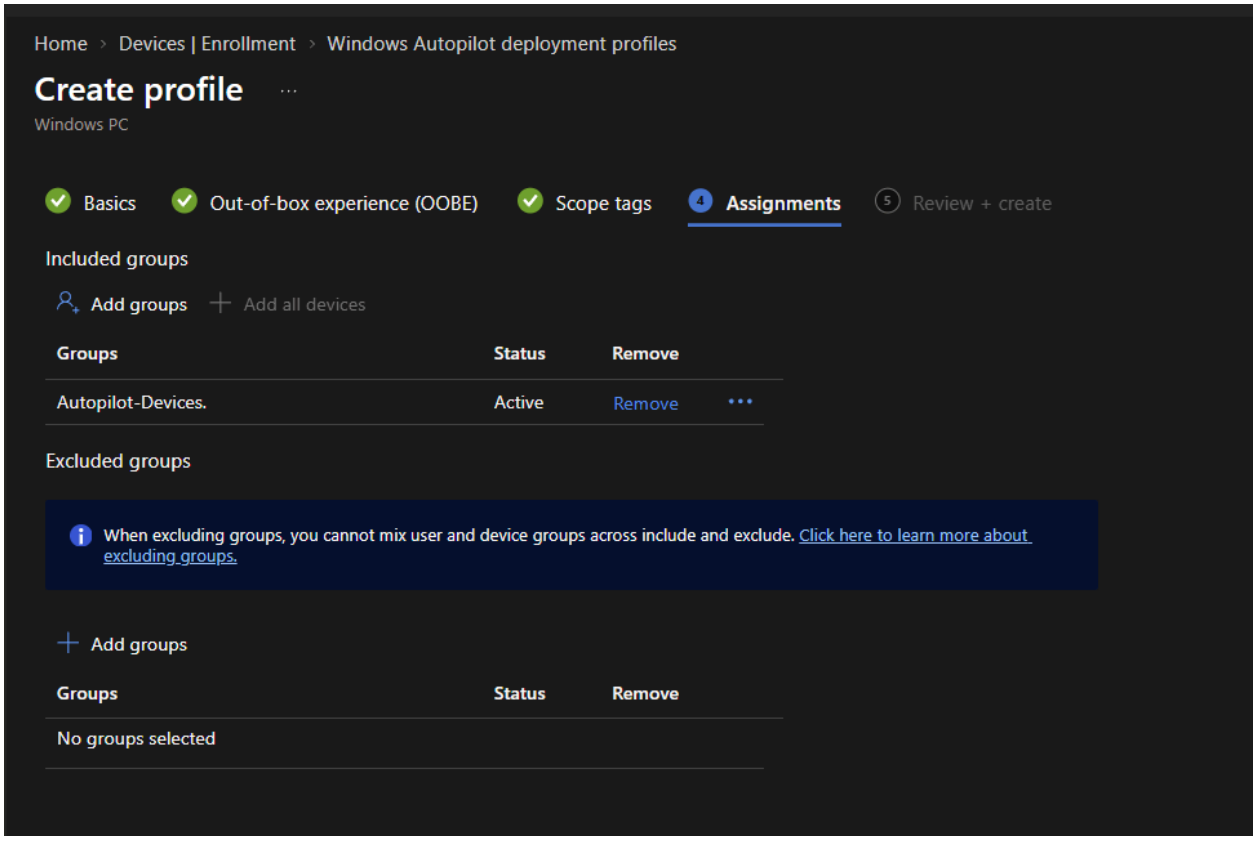
1. In the **Microsoft Intune admin center**, go to **Devices → Enrollment → Windows Autopilot → Deployment profiles**.
2. Select **Create profile → Windows PC**.
3. **Basics:** Enter a name, for example `AutopilotUserDrivenProfile`, and an optional description.
4. **Convert all targeted devices to Autopilot: Yes :** In this lab, the device was registered by capturing its hardware hash. Turning this setting on means that if we later add devices that are already enrolled in Intune but not yet registered in Autopilot, they're registered automatically instead of requiring a hash capture for each one. The deployment profile then applies the next time the device goes through OOBE, such as after a reset.



5. **Out-of-box experience (OOBE):** Configure the settings:
    - Deployment mode: **User-driven**
    - Join to Microsoft Entra ID as: **Microsoft Entra joined**
    - Microsoft Software License Terms: **Hide**
    - Privacy settings: **Hide**
    - Hide change account options: **Hide**
    - User account type: **Standard**
    - Allow pre-provisioned deployment**: No**
    - Optional: apply a device name template, for example `NYC-%RAND:5%`



6. **Assignments:** Under **Included groups**, select **Add groups** and choose the dynamic group we just created.



7. **Review + create**, then select **Create**.

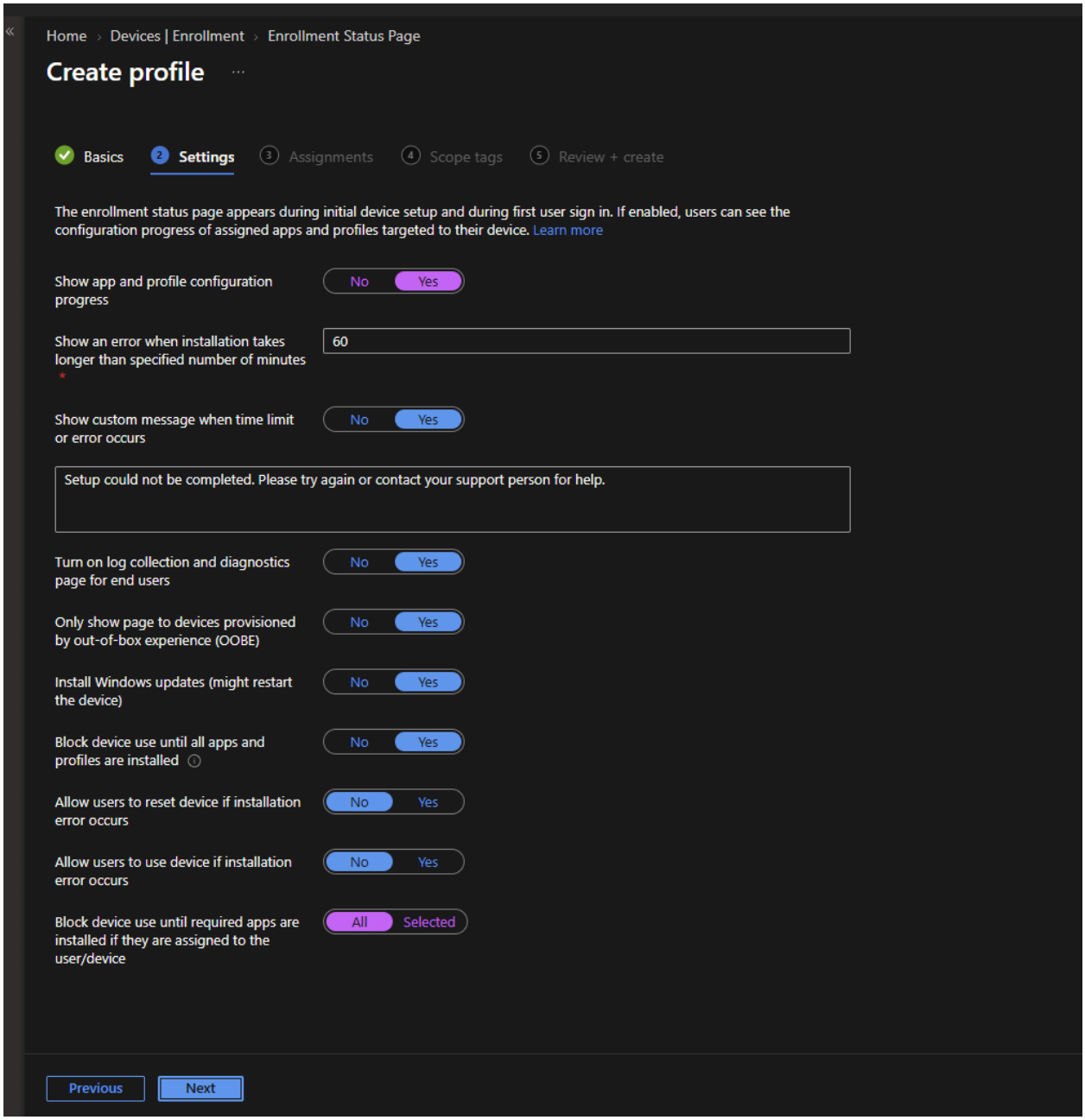
Create an Enrollment Status Page (ESP)

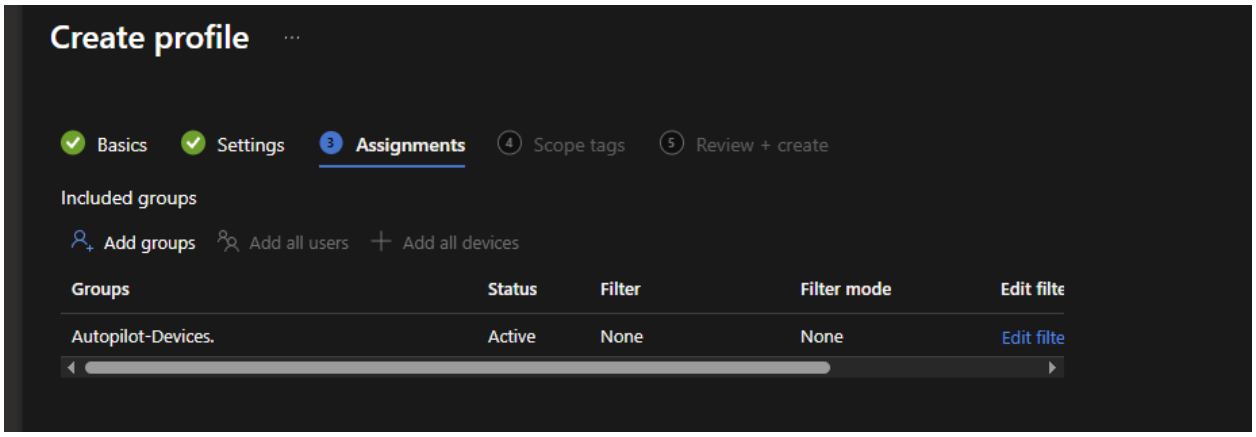
The Enrollment Status Page is the progress screen users see during Autopilot setup. It tells the user that the device is still being configured and tracks the apps, profiles, and policies being installed. It can also **block the user from reaching the desktop** until the required items finish, so the device is fully configured on first use.

1. In the **Microsoft Intune admin center**, go to **Devices → Enrollment → Windows Autopilot → Enrollment Status Page**
2. Select **Create**.
3. **Basics:** Enter a name, for example `Autopilot-ESP`.
4. **Settings:** Set **Show app and profile configuration progress** to **Yes**, then configure the options that appear:
    - Show an error when installation takes longer than the specified number of minutes
    - Show custom message when time limit or error occurs
    - Allow users to collect logs about installation errors

- Only show page to devices provisioned by out-of-box experience (OOBE)
- Block device use until all apps and profiles are installed
- Allow users to reset device if an installation error occurs
- Allow users to use device if an installation error occurs

5. **Assignments:** Assign it to your dynamic group.
6. Select **Create**





Test the deployment in OOBE

The hardware hash is registered, the dynamic group contains the device, and the deployment profile is assigned to that group. Next, take the device to the OOBE (out-of-box experience) to test it.

What happens:

1. The device boots into OOBE and connects to the internet.
2. It contacts Microsoft's Autopilot service and asks who it belongs to.
3. Microsoft checks the device's hardware hash and identifies which organization (tenant) it's registered to.
4. The device downloads the Autopilot deployment profile assigned to it from that tenant.
5. The device shows the organization's sign-in screen with company branding and asks the user to authenticate.
6. After authentication, OOBE continues as the profile defines (which pages are hidden or shown, user account type, and so on).
7. The device is joined to Microsoft Entra ID and enrolled in Intune.
8. The Enrollment Status Page (ESP) shows progress while Intune applies the configuration, app, and security policies assigned to the device or user.

Return a device to OOBE (if it's already set up)

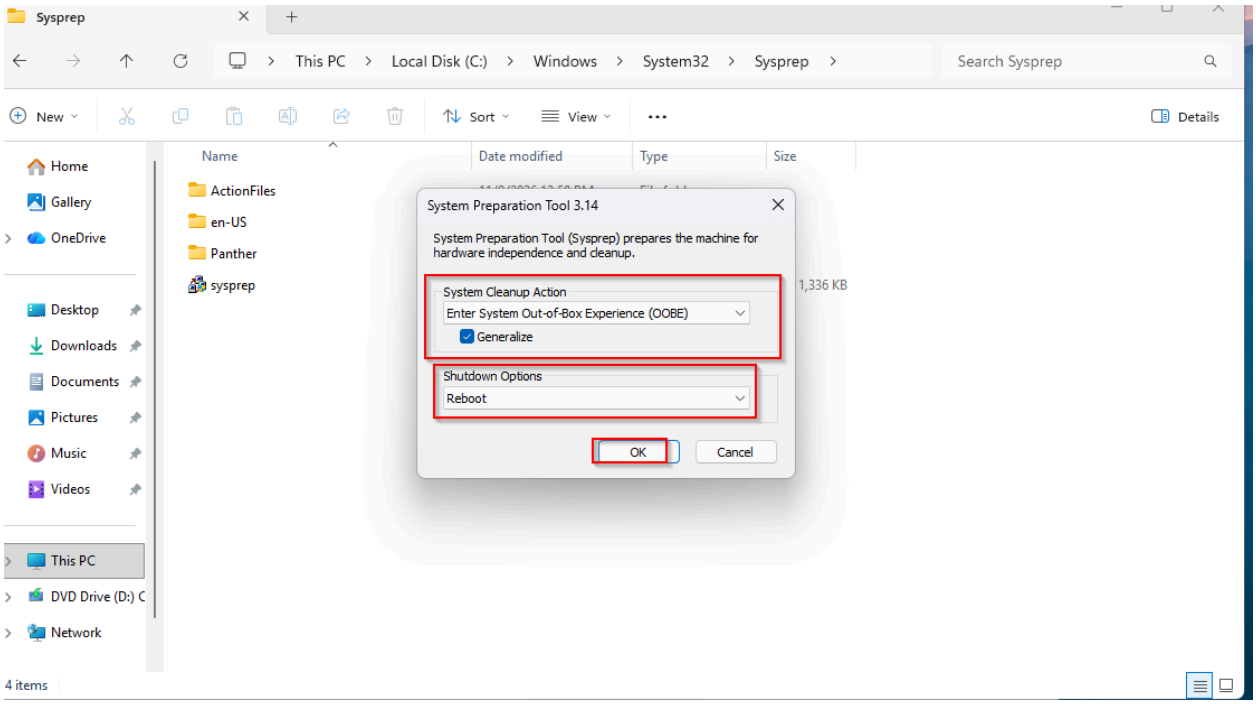
If the device has already been through OOBE and is sitting at the Windows desktop, you can send it back to OOBE with **Sysprep**, a tool built into Windows (`C:\Windows\System32\Sysprep\sysprep.exe`).

1. Open Sysprep. Run `C:\Windows\System32\Sysprep\sysprep.exe`, or use the command line below.

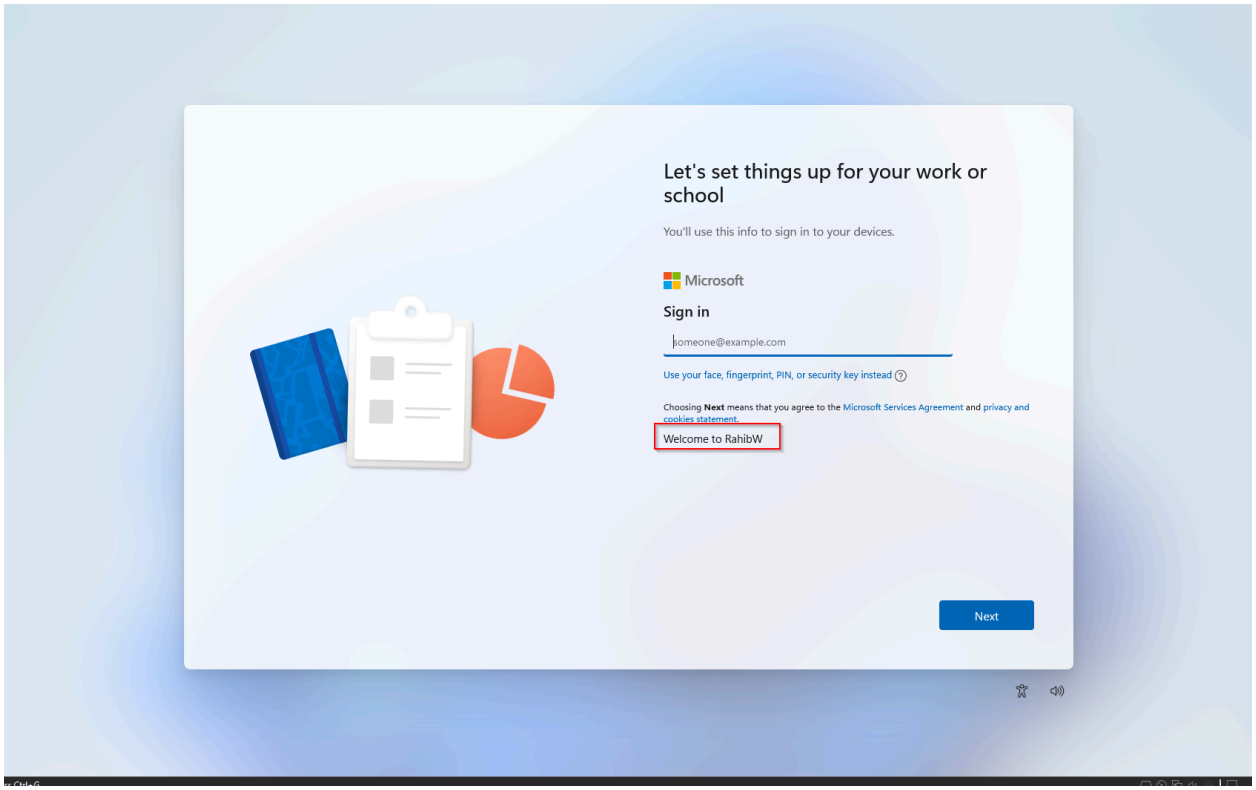
2. Set **System Cleanup Action** to **Enter System Out-of-Box Experience (OOBE)**.
3. Check **Generalize**.
4. Set **Shutdown Options** to **Reboot** (or **Shutdown**).
5. Select **OK**. When the device starts again, it goes to OOBE.

Command-line version:

```
C:\Windows\System32\Sysprep\sysprep.exe /oobe /generalize /reboot
```



Now we are in OOBE:



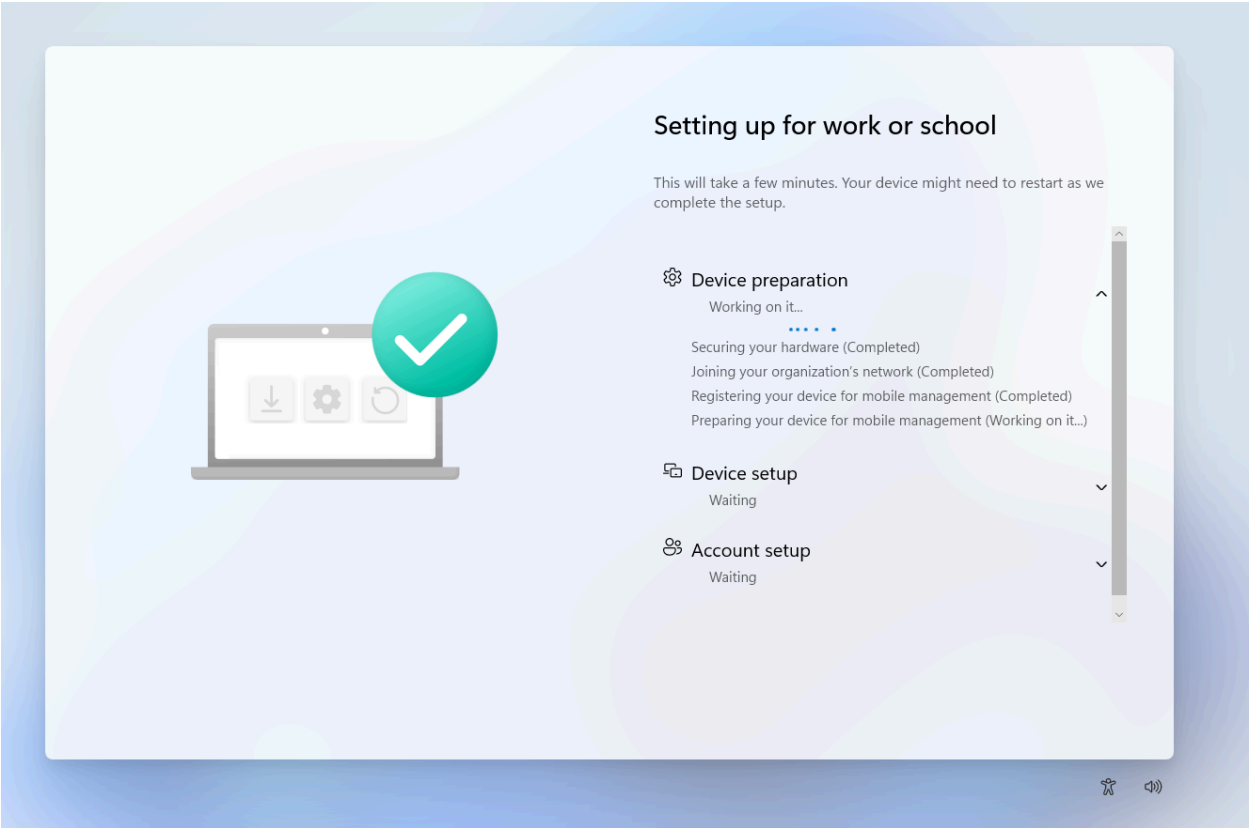
At the sign-in screen during OOBE, the organization's company branding is already displayed, even though no credentials have been entered. The device identified its tenant because its hardware hash was registered in Autopilot. On startup, it contacted Microsoft, which matched the hash to our tenant, and the device then loaded the tenant's sign-in page with its branding.

There is also no option to create a local account. The user is taken straight to the work sign-in screen and must authenticate with an organizational account. This follows from the Autopilot deployment profile, which forces the device to join Microsoft Entra ID.

Sign in and watch the device setup

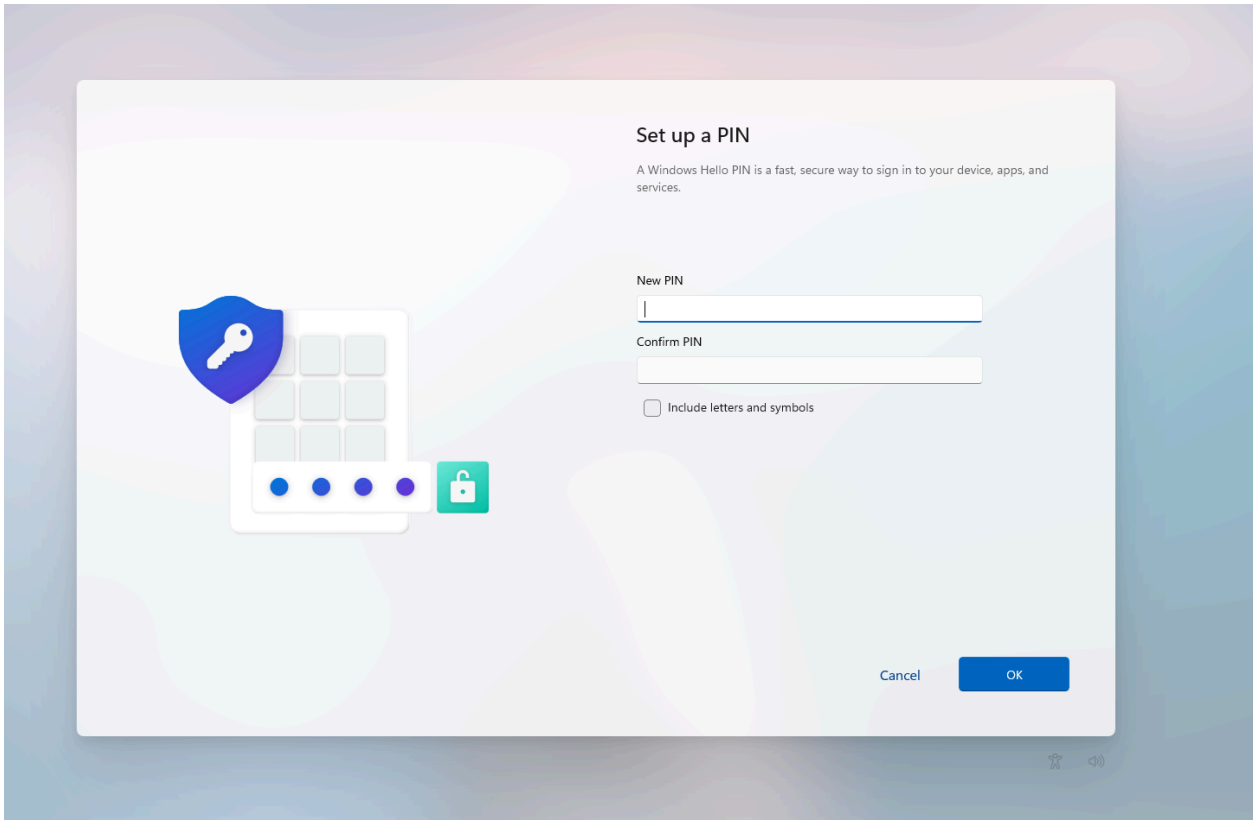
After signing in, the device shows the **Enrollment Status Page** with the progress of the device being set up. This page appears because we created and assigned an Enrollment Status Page profile.

The user only had to sign in. They weren't shown the license agreement (EULA), the privacy settings, or the account setup pages, because the Autopilot deployment profile hides them.



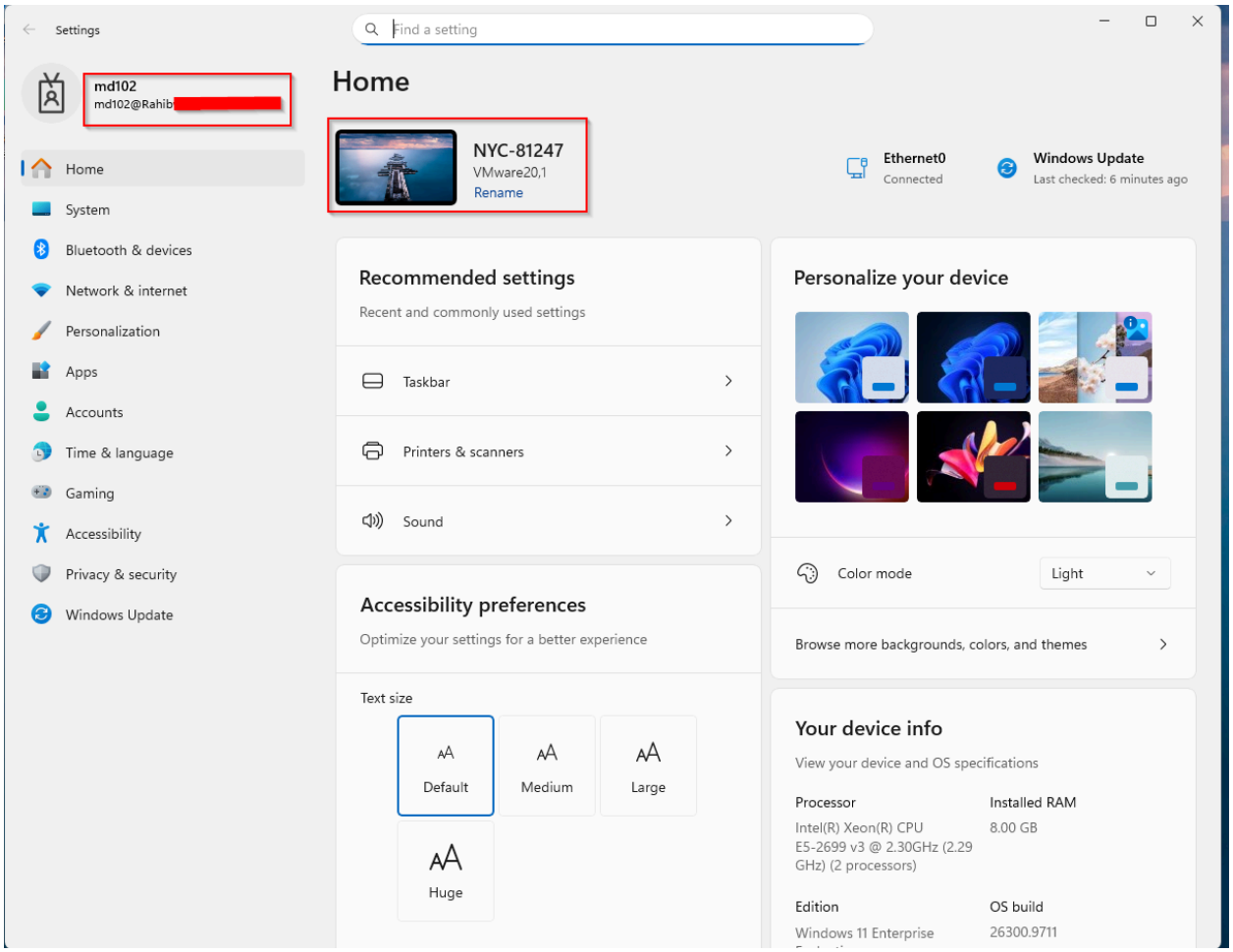
Windows Hello for Business (PIN setup)

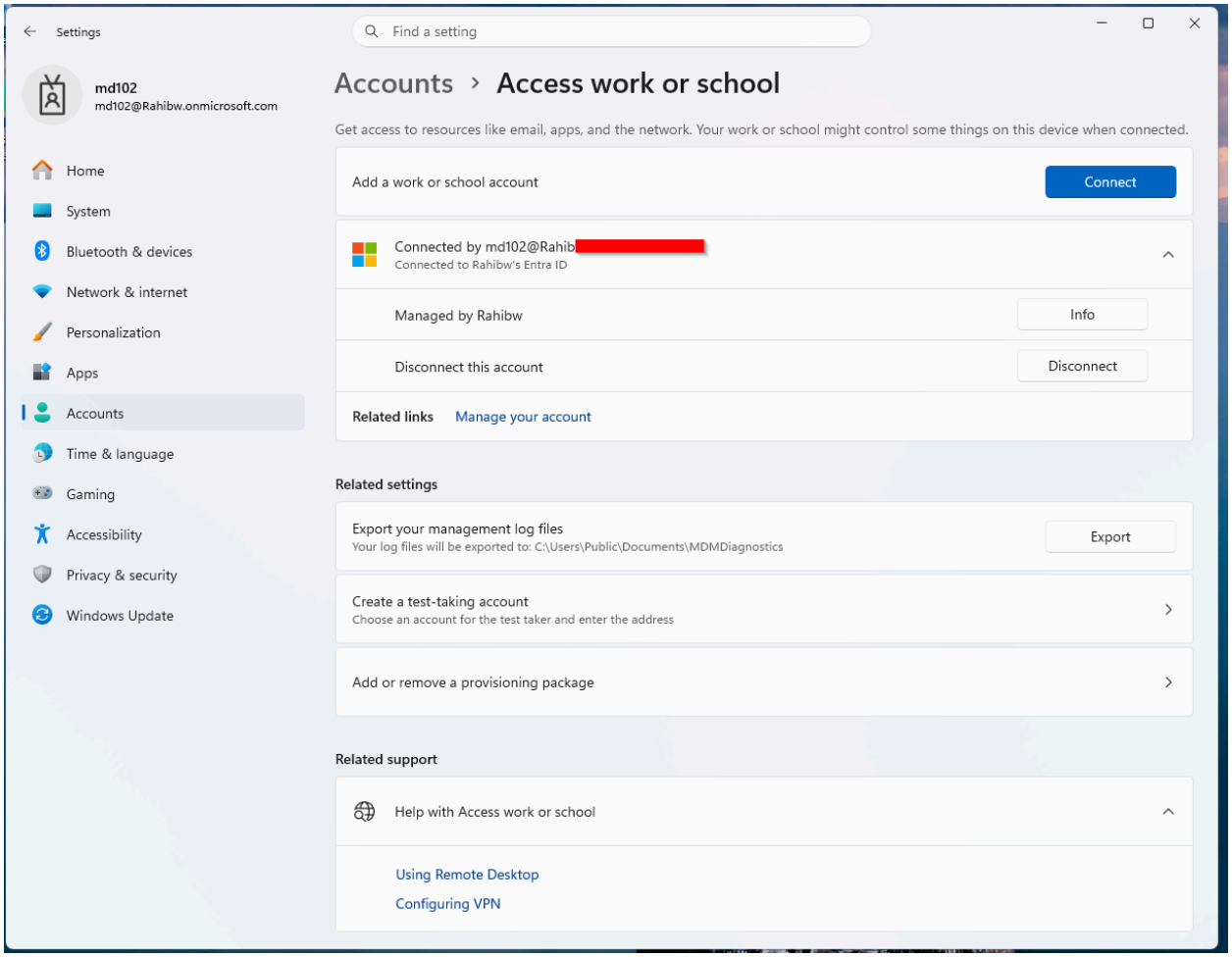
If a Windows Hello for Business policy is assigned to the user or device, the user is also prompted to create a PIN during setup. The policy decides whether a PIN is required, and the user sets it on this machine before reaching the desktop.



Result: the device is ready

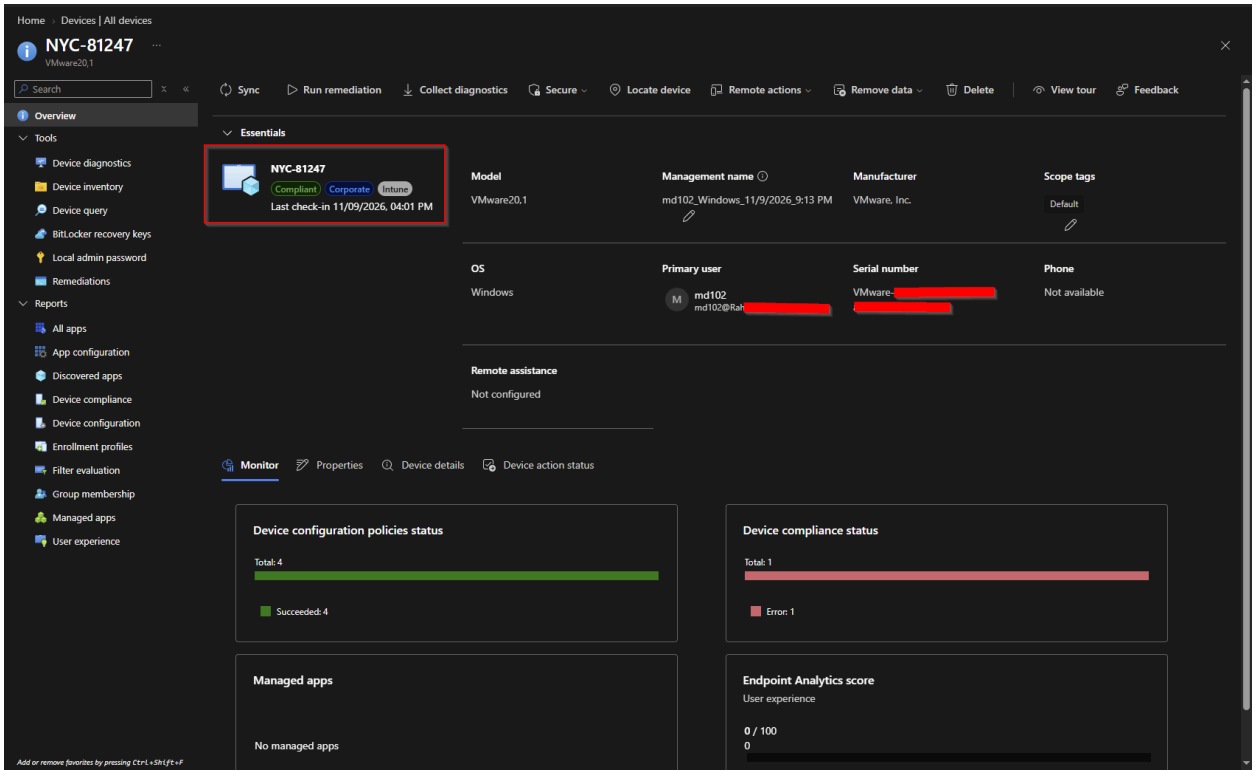
Setup is finished and the device reached the desktop. It also followed the naming template. The device is named **NYC-81247** because the template configured in the deployment profile was `NYC-%RAND:5%` , which produces the prefix `NYC-` followed by a random 5-digit number.

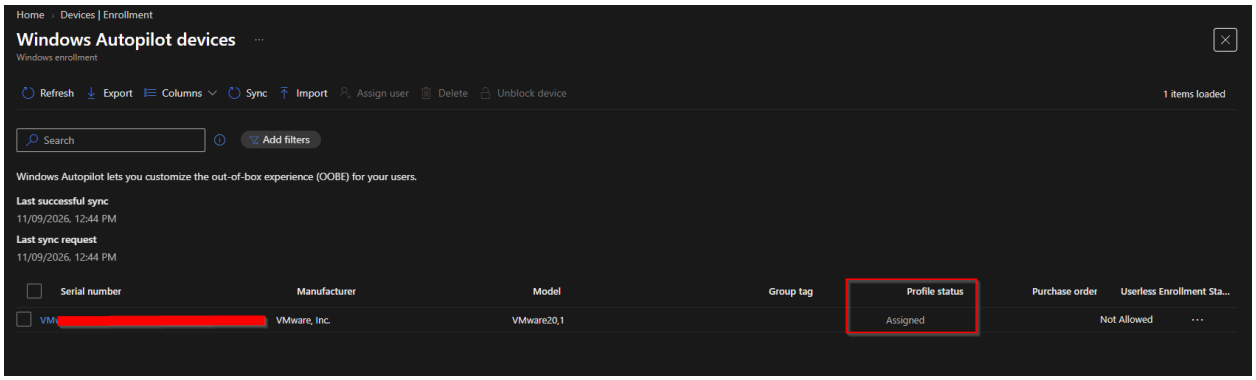

Settings
Find a setting
md102
md102@Rahib
Home
System
Bluetooth & devices
Network & internet
Personalization
Apps
Accounts
Time & language
Gaming
Accessibility
Privacy & security
Windows Update
Home
NYC-81247
VMware20,1
Rename
Ethernet0
Connected
Windows Update
Last checked: 6 minutes ago
Recommended settings
Recent and commonly used settings
Taskbar
Printers & scanners
Sound
Accessibility preferences
Optimize your settings for a better experience
Text size
Default
Medium
Large
Huge
Personalize your device
Color mode
Light
Browse more backgrounds, colors, and themes
Your device info
View your device and OS specifications
Processor
Intel(R) Xeon(R) CPU E5-2699 v3 @ 2.30GHz (2.29 GHz) (2 processors)
Installed RAM
8.00 GB
Edition
Windows 11 Enterprise
OS build
26300.9711



Result in Intune: the device is enrolled

In the Microsoft Intune admin center, under **Devices** → **All devices**, the device now appears as **NYC-81247**. This confirms it enrolled and is managed by Intune. The device is listed with the primary user who signed in during OOBE, and its Autopilot deployment profile shows as assigned.


Home › Devices | All devices
NYC-81247
VMware20,1
Search
Overview
Tools
Device diagnostics
Device inventory
Device query
BitLocker recovery keys
Local admin password
Remediations
Reports
All apps
App configuration
Discovered apps
Device compliance
Device configuration
Enrollment profiles
Filter evaluation
Group membership
Managed apps
User experience
Add or remove favorites by pressing Ctrl+Shift+F
Sync
Run remediation
Collect diagnostics
Secure
Locate device
Remote actions
Remove data
Delete
View tour
Feedback
Essentials
NYC-81247
Compliant
Corporate
Intune
Last check-in 11/09/2026, 04:01 PM
Model
VMware20,1
Management name
md102_Windows_11/9/2026_9:13 PM
Manufacturer
VMware, Inc.
Scope tags
Default
OS
Windows
Primary user
md102
Serial number
Phone
Not available
Remote assistance
Not configured
Monitor
Properties
Device details
Device action status
Device configuration policies status
Total: 4
Succeeded: 4
Device compliance status
Total: 1
Error: 1
Managed apps
No managed apps
Endpoint Analytics score
User experience
0 / 100
0


Home › Devices | Enrollment
Windows Autopilot devices
Windows enrollment
Refresh
Export
Columns
Sync
Import
Assign user
Delete
Unblock device
1 items loaded
Search
Add filters
Windows Autopilot lets you customize the out-of-box experience (OOBE) for your users.
Last successful sync
11/09/2026, 12:44 PM
Last sync request
11/09/2026, 12:44 PM
Serial number
Manufacturer
Model
Group tag
Profile status
Purchase order
Userless Enrollment Sta...
VMware, Inc.
VMware20,1
Assigned
Not Allowed